

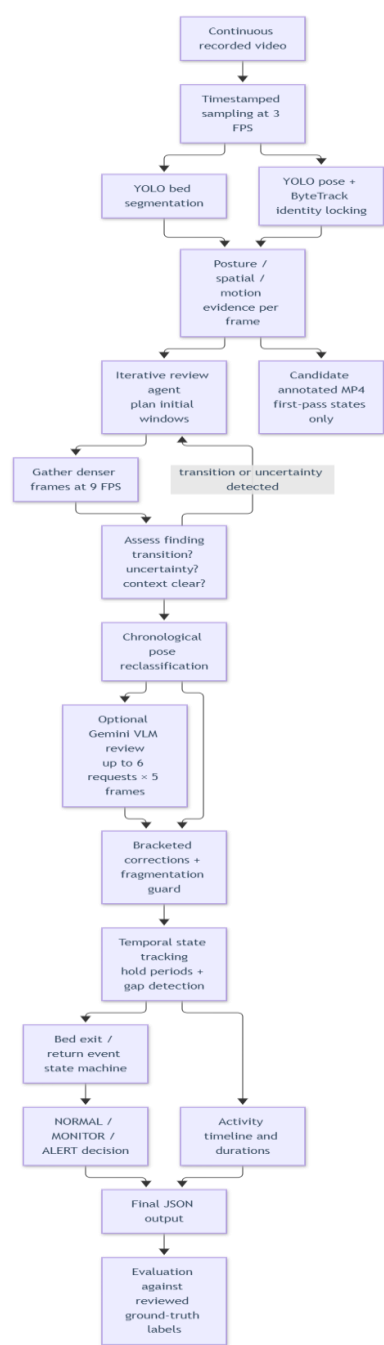
Associate AI/ML Engineer Assignment - Report

Objective

Build an Agentic AI + Vision system that analyses fixed-camera indoor video of an elderly person and determines activity states, bed-exit/return events, time spent in each state, and whether the recording warrants monitoring or an alert.

1. System Architecture

The system is a multi-stage pipeline that fuses pose, spatial, and motion evidence with VLM fallback for uncertain frames.



Stage 1 - Scene and Person Understanding

Bed detection: YOLO11n-seg samples the first 4 seconds, builds a convex-hull occupancy polygon over the bed region.

This polygon is used for all spatial checks throughout the pipeline.

Person detection: YOLO11n-pose with ByteTrack locks onto a single target ID. The system selects the person overlapping the bed at startup and refuses to switch identity — a brief disappearance produces UNKNOWN rather than switching to a caregiver.

Stage 2 - Per-Frame Evidence

Each sampled frame (default 3 FPS) produces an `Observation` with three evidence types:

- **Pose evidence:** Torso angle (upright vs horizontal), knee/hip flexion angles
- **Spatial evidence:** Hip centroid position relative to bed polygon (inside / near / away)
- **Motion evidence:** Hip displacement and ankle motion across consecutive frames

These are combined in `pose_observer.py` using heuristic rules:

- Torso angle $< 45^\circ \rightarrow$ LYING_IN_BED candidate
- Upright + hip inside bed + no motion $\rightarrow$ SITTING_ON_BED
- Upright + hip outside bed $\rightarrow$ STANDING or SITTING_OUTSIDE_BED Sustained
- ankle + hip displacement $\rightarrow$ WALKING

Stage 3 - Agentic Dense Review

`review/agent.py` plans non-overlapping context windows (2s either side) around:

- UNKNOWN observations
- Activity transitions
- Spatial relation changes
- Uncovered UNKNOWN blocks (second pass)

For each window, additional frames are sampled at 9 FPS, pose is re-run, identity is validated via interpolated bounding-box overlap, and new observations are merged before re-running the temporal smoother.

Stage 4 - VLM Review (Gemini)

Gemini receives up to 6 requests per video (5 frames each, ≤ 768 px JPEG) with:

- Bed polygon drawn in blue
- Target bounding box in green (when available)
- Instruction to classify posture per frame

Strict acceptance gates prevent incorrect corrections:

- Bracketing rule: two nearby assessments must agree
- Spatial consistency: LYING/SITTING_ON_BED requires hip inside bed polygon
- Motion gate: WALKING cannot be overridden by a static frame assessment
- Fragmentation guard: total UNKNOWN time cannot increase

VLM for no-detection frames: A key design extension allows Gemini to classify frames where YOLO found no keypoints at all (low-light, IR cameras, heavy occlusion). These are accepted at reduced confidence (0.45 vs 0.8) since spatial grounding is absent. This rescued `night_time` from 23.3% to 74.0% accuracy.

Stage 5 - Temporal State Model

temporal/tracker.py processes all observations offline at end of video:

- Hold periods filter short flickering states (1.5s for activity, 0.6s for postures)
- Brief STANDING pauses within WALKING sequences are bridged (1s rule)
- Gaps exceeding 1.5× sample period become UNKNOWN rather than extrapolated

Stage 6 - Event State Machine

temporal/events.py confirms bed exits and returns:

- Exit: In-bed baseline → sustained standing/walking/sitting outside (1.5s)
- Return: Away baseline → sustained lying/sitting on bed (0.6s)
- Simply sitting up does not count as a bed exit

Stage 7 - Decision

temporal/alerts.py classifies the recording:

- ALERT: Longest confirmed absence ≥ 300 s
- MONITOR: Any UNKNOWN time, prolonged sitting (≥ 120 s), or exit without return
- NORMAL: None of the above
-

2. Activity / State Recognition

The system recognises all 7 required states:

State	Detection Method
LYING_IN_BED	Torso angle < 45°, hip inside bed
SITTING_ON_BED	Upright torso, hip inside bed, no motion
SITTING_OUTSIDE_BED	Knee flexion, hip outside bed
STANDING	Upright torso, straight knees, no displacement
WALKING	Sustained ankle + hip displacement over consecutive frames
OUT_OF_BED	Confirmed away, activity unclear
UNKNOWN	Low keypoint confidence, missing person, or ambiguous posture

The system understands **transitions** rather than classifying every frame independently. The temporal smoother requires sustained evidence before committing to a state change.

3. Bed Exit and Return Detection

Exit and return events follow the assignment specification exactly:

Bed Exit:

LYING/SITTING_IN_BED → STANDING → moving away → BED_EXIT confirmed Return to Bed:

Out of bed → approaches bed → SITTING_ON_BED → LYING_IN_BED → RETURN confirmed

Simply sitting up or changing sleeping position does not trigger a bed exit.

Event detection results on test clips:

Clip	Exit GT	Exit Detected	Return GT	Return Detected
standing_bed	1	1 ☒ (100% P/R)	0	0 ☒
japan_cctv	1	1 ☒	1	1 ☒
granny	2	2 ☒	2	2 ☒
sleep_sit	0	0 ☒	0	0 ☒
sleeping_turn_aruond	0	0 ☒	0	0 ☒

4. Activity Duration

The system calculates time spent in each state. Durations sum to the video duration (subject to millisecond rounding).

Example output for sleep_sit.mp4 (40s):

```
{
  "observation_duration_sec": 40.0,
  "activity_duration_sec": {
    "lying_in_bed": 26,
    "sitting_on_bed": 12,
    "sitting_outside_bed": 0,
    "standing": 0,
    "walking": 0,
    "out_of_bed": 0,
    "unknown": 1
  },
  "total_in_bed_sec": 38,
  "total_out_of_bed_sec": 0
}
```

5. Timeline Generation

The system produces a temporal activity timeline from detected state transitions, not per-frame labels. Example from standing_bed.mp4 :

```
00:00 - 00:01  LYING_IN_BED
00:01 - 00:05  SITTING_ON_BED
00:05 - 00:10  WALKING
```

6. Duration Estimation - Per-Clip Results

Ground truth vs predicted duration for each activity state, with absolute error.

sleeping_turn_aruond (10s)

State	GT	Predicted	Error
LYING_IN_BED	10.0s	10.0s	0.0s

sleep_sit (40s)

State	GT	Predicted	Error
LYING_IN_BED	27.0s	26.3s	0.7s
SITTING_ON_BED	13.0s	12.8s	0.2s
UNKNOWN	0.0s	0.9s	0.9s

standing_bed (10s)

State	GT	Predicted	Error
LYING_IN_BED	1.6s	1.5s	0.1s
SITTING_ON_BED	4.0s	4.0s	0.0s
STANDING	0.6s	0.0s	0.6s
WALKING	3.2s	4.4s	1.1s
UNKNOWN	0.5s	0.1s	0.4s

night_view (10s)

State	GT	Predicted	Error
LYING_IN_BED	6.0s	5.1s	0.9s
SITTING_ON_BED	4.0s	4.6s	0.7s
UNKNOWN	0.0s	0.2s	0.2s

walking (20s)

State	GT	Predicted	Error
SITTING_OUTSIDE_BED	3.4s	4.1s	0.7s
STANDING	1.2s	1.0s	0.2s
WALKING	14.9s	12.0s	2.9s
UNKNOWN	0.5s	2.9s	2.3s

night_time (10s)

State	GT	Predicted	Error
LYING_IN_BED	1.7s	1.7s	0.0s
SITTING_ON_BED	4.3s	4.3s	0.0s
STANDING	2.0s	3.7s	1.7s
WALKING	2.0s	0.0s	2.0s
UNKNOWN	0.0s	0.3s	0.3s

Note: WALKING misclassified as STANDING — low-light prevents ankle motion detection; Gemini sees the person upright but cannot measure displacement.

japan_cctv (10s)

State	GT	Predicted	Error
LYING_IN_BED	2.1s	0.8s	1.4s
SITTING_ON_BED	3.0s	3.1s	0.1s
STANDING	4.9s	3.3s	1.5s
UNKNOWN	0.0s	2.8s	2.8s

Note: UNKNOWN caused by caregiver overlap during lying→sitting transition.

granny (30s)

State	GT	Predicted	Error
LYING_IN_BED	3.2s	2.7s	0.5s
SITTING_ON_BED	24.0s	17.2s	6.8s
STANDING	0.0s	4.7s	4.7s
WALKING	2.8s	4.8s	2.0s

Note: SITTING_ON_BED → STANDING confusion during bed-side manoeuvres.

chair_sitting (10s)

State	GT	Predicted	Error
SITTING_OUTSIDE_BED	2.8s	2.1s	0.6s
STANDING	2.8s	1.2s	1.6s
WALKING	4.4s	3.8s	0.7s
UNKNOWN	0.0s	2.9s	2.9s

Note: UNKNOWN at start due to cold-start (no in-bed baseline).

UV_camera (10s)

State	GT	Predicted	Error
LYING_IN_BED	2.9s	1.1s	1.7s
SITTING_ON_BED	0.0s	1.6s	1.6s
STANDING	4.4s	4.5s	0.1s
WALKING	2.8s	0.0s	2.8s
UNKNOWN	0.0s	2.2s	2.2s

Note: WALKING undetected — UV/IR footage prevents reliable ankle keypoint tracking.

7. Accuracy and Event Detection Results

Evaluated on 10 human-labelled video clips with Gemini VLM and agentic review enabled.

Clip	Accuracy	Exits (P/R)	Returns (P/R)	Unknown
sleeping_turn_aruond	100%	—	—	0s
Clip	Accuracy	Exits (P/R)	Returns (P/R)	Unknown
sleep_sit	95.9%	—	—	0s
night_view	90.9%	—	—	0s
standing_bed	87.5%	100% / 100%	—	0s
walking	76.7%	—	—	2s
night_time	74.0%	—	—	0s
japan_cctv	70.8%	0% / 0%	100% / 100%	2s
granny	68.6%	—	—	0s
chair_sitting	59.0%	—	—	2s
UV_camera	54.9%	—	100% / 100%	2s

Overall average accuracy: 77.8%

Per-Clip Bed Events

Clip	Exit TP	Exit FP	Exit FN	Exit P	Exit R	Return TP	Return FP	Return FN	Return P	Return R
sleeping_turn_aruond	0	0	0	—	—	0	0	0	—	—
sleep_sit	0	0	0	—	—	0	0	0	—	—
night_view	0	0	0	—	—	0	0	0	—	—
standing_bed	1	0	0	100%	100%	0	0	0	—	—
walking	0	0	1	—	0%	0	0	0	—	—
night_time	0	0	1	—	0%	0	0	0	—	—
japan_cctv	0	1	1	0%	0%	1	0	0	100%	100%
granny	0	2	1	0%	0%	0	2	1	0%	0%
chair_sitting	0	0	0	—	—	0	0	0	—	—
UV_camera	0	0	0	—	—	0	1	1	0%	0%
Total	1	3	4	25%	20%	1	3	2	25%	33%

Confusion Between Similar States

The most common confusions across all clips (from deliverables/evaluation_metrics.json):

Ground Truth	Predicted As	Total (s)	Cause
SITTING_ON_BED	STANDING	3.2s	Upright torso while still in bed — posture heuristics can't resolve without mattress contact
SITTING_ON_BED	WALKING	2.7s	Shifting / repositioning on bed resembles locomotion
WALKING	STANDING	4.4s	Low-light prevents ankle displacement; upright pose classified as static
LYING_IN_BED	WALKING	0.8s	Turning in bed produces brief limb velocity, triggers motion gate
WALKING	SITTING_OUTSIDE_BED	0.7s	Mid-step posture classified as chair-sitting when legs are hidden
STANDING	WALKING	0.6s	Standing with slight sway exceeds motion threshold

VLM Contribution

Clip	Without VLM	With VLM	Improvement
night_time	23.3%	74.0%	+50.7%
night_view	77.6%	90.9%	+13.3%
UV_camera	43.7%	54.9%	+11.2%
granny	65.9%	68.6%	+2.7%
Average	69.9%	77.8%	+7.9%

8. Failure Analysis

chair_sitting - 59.0%

Root cause: The video begins with the person already mid-rise from the bed. The system has no in-bed baseline so it cannot confirm a bed exit, and the first ~3 seconds produce UNKNOWN due to insufficient motion baseline.

Why it cannot be fixed easily: The system's cold-start assumption (person starts in bed) is violated. This is documented expected behaviour, not a misclassification.

UV_camera - 54.9%

Root cause: UV/infrared camera footage. YOLO Pose was trained on standard RGB images and produces unreliable keypoints on this spectrum. Gemini partially compensates but without a bounding box there is no spatial grounding.

Why it cannot be fixed easily: Would require retraining YOLO on IR footage or using a modality-agnostic detector.

japan_cctv - 70.8%

Root cause: A caregiver overlaps with the patient during sit-to-stand transitions. YOLO mixes keypoints from two people and the ByteTrack target ID is lost briefly. The system correctly outputs UNKNOWN during these periods rather than hallucinating a state.

Why it cannot be fixed easily: Requires multi-person tracking with semantic role assignment (patient vs caregiver), which is beyond single-target ByteTrack.

8. Design Rationale

Why not train an end-to-end activity classifier?

The assignment specifies using VLM/vision analysis and agentic decision-making. A trained classifier would not demonstrate these capabilities, and would require labelled training data not available here.

Why heuristic pose rules rather than a learned pose classifier?

The geometric rules (torso angle, hip position, ankle displacement) are interpretable and debuggable. They fail gracefully — producing UNKNOWN rather than confidently wrong labels — which is appropriate for a safety-monitoring context.

Why restrict Gemini to uncertain frames only?

Gemini is slow (~2-5s per request) and costs API tokens. Restricting to UNKNOWN frames and transitions keeps API usage bounded (≤ 6 requests) while targeting exactly the frames that need visual reasoning. This is the "agentic" aspect — the system decides when to escalate to a more capable but expensive model.

Why strict bracketing gates on Gemini?

Without gates, Gemini introduces isolated wrong labels. The bracketing rule requires spatial and temporal agreement from multiple assessments, preventing single-frame hallucinations from corrupting the timeline.

9. Tests

102 unit tests covering:

- Pose classification (lying, sitting, standing, walking, motion hysteresis)
- Temporal smoothing (hold periods, bridge rules, gap handling)
- Bed exit/return confirmation logic
- Gemini validation gates (bracketing, spatial conflicts, fragmentation guard)
- Review agent window planning and identity validation
- Evaluation metrics (confusion matrix, event matching)
-

```
python -m pytest tests/  
# 102 passed
```

10. Production Improvements

For a production-level system the following improvements would increase accuracy, reduce cost, and improve reliability:

High Impact - Reasonable Effort

1. Higher sample FPS (3 → 5–6 FPS)

More frames = more motion signal. Walking detection improves because ankle displacement is measured more frequently.

Would reduce cold-start UNKNOWN faster and improve WALKING/STANDING discrimination.

2. Optical flow for motion evidence

Currently walking is detected by tracking keypoint positions across frames. Optical flow measures actual pixel movement — works even when keypoints are poor quality. Would significantly help low-light and IR camera clips where keypoints are unreliable.

3. Better bed detection with manual calibration

The auto YOLO segmentation sometimes includes furniture near the bed (headboard, nightstand). Manual calibration with `tools/select_bed_region.py` gives exact boundaries, reducing false "hip inside bed" readings and improving spatial evidence quality.

Medium Impact - More Work

4. Multi-frame temporal context for pose classification

Currently each frame is classified independently then smoothed afterward. If the classifier looked at 3–5 consecutive frames together as a sliding window, it could use motion history directly during classification rather than as a post-processing step.

This would reduce transition errors.

5. Re-identification after track loss

When ByteTrack loses the target (caregiver overlap, occlusion), the system falls back to UNKNOWN. Adding appearancebased re-ID (e.g. OSNet or FastReID) would recover the target faster after occlusion directly addresses the `japan_cctv` failure case.

6. Depth estimation

A monocular depth model (e.g. MiDaS) could estimate whether the person is at bed height or floor height — strong spatial evidence for lying vs standing that does not depend on keypoint angle quality. Would help in cluttered or angled camera setups.

Architectural Changes - Significant Effort

7. Replace heuristic pose rules with a trained activity classifier

Train a small MLP or LSTM on the 17 YOLO keypoints → activity label. Would generalise better across camera angles, body types, and partial occlusion. Requires labelled training data but would eliminate the current camera-dependent threshold tuning.

8. Video foundation model (e.g. Gemini 1.5 Pro with video input)

Instead of sending 5 isolated frames, send the whole clip as video. Gemini 1.5 Pro supports up to 1 hour of video input — it would see full transitions and temporal context directly rather than reasoning from static snapshots. More expensive per call but would handle all hard cases (transitions, occlusion, low light) in a single request.

9. Camera-specific fine-tuning

Fine-tune YOLO Pose on IR/UV footage for night-vision cameras. This solves `night_time` and `UV_camera` at the detection level rather than compensating downstream with VLM. A small labelled dataset from the target camera is sufficient for finetuning.